Understanding Missing Values

by Shahriar Analytical Academy

What Are Missing Values?

Definition: Missing values are data entries that are not stored or are absent in a dataset. This means that for certain records, one or more values are unknown, blank, or undefined.

Why It Happens:



The respondent skipped a question in a survey



A sensor failed to record a reading



Data was lost due to a system error



The value doesn't apply to a particular case (e.g., age of an unborn child)

Real-World Example: Imagine you're analyzing customer feedback. One customer didn't write their age or location. Those are missing values, and they need to be handled before analysis.

Types of Missing Data

There are **three main types** of missing data. Understanding the type helps decide how to handle it.

1 MCAR (Missing Completely at Random)

The missingness is independent of any values in the dataset. Example: A survey form is damaged randomly.

2 MAR (Missing at Random)

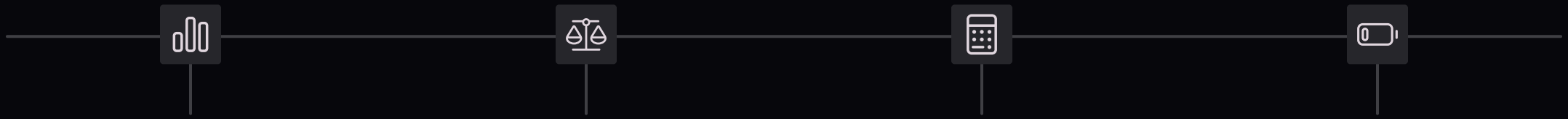
The missingness depends on observed data but not on the missing value itself. Example: Income is missing more often for younger people.

3 MNAR (Missing Not at Random)

The missingness is related to the missing value itself. Example: People with high income may choose not to reveal it.

How Missing Values Affect Data Analysis

Consequences:



Reduces Data Quality

Skewed or incomplete insights.

Biases Results

Especially if the missing data has a pattern.

Statistical Errors

Many statistical tools and machine learning algorithms cannot handle missing values directly.

Loss of Power

Less data means less confidence in findings.

Example: If you drop all rows with missing values from a 1000-entry dataset and are left with only 400 entries, the results may not represent the full population anymore.

Summary and Key Learning Points

1

Missing Values Impact

Missing values are common and can distort analysis if ignored.

2

Types Matter

They come in different types: MCAR, MAR, and MNAR — understand them before handling.

3

Explore First

Always explore your data visually and statistically to detect missingness patterns.

4

Strategy Selection

Choose your handling strategy wisely: deletion, imputation, or flagging.

Thank You!

Shahriar Analytical Academy